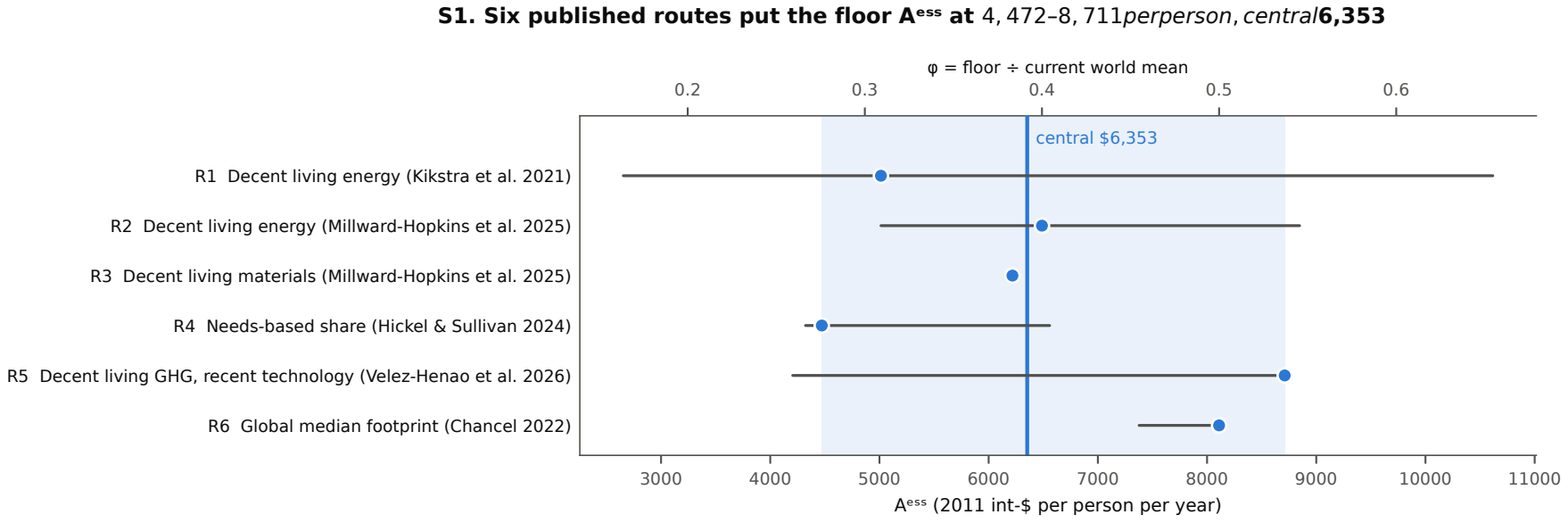
S1. Six published routes put the floor Aess at 4,472–8,711 per person, central 6,353
φ = floor ÷ current world mean
0.2
0.3
0.4
0.5
0.6
central $6,353
R1 Decent living energy (Kikstra et al. 2021)
R2 Decent living energy (Millward-Hopkins et al. 2025)
R3 Decent living materials (Millward-Hopkins et al. 2025)
R4 Needs-based share (Hickel & Sullivan 2024)
R5 Decent living GHG, recent technology (Velez-Henao et al. 2026)
R6 Global median footprint (Chancel 2022)
3000
4000
5000
6000
7000
8000
9000
10000
11000
Aess (2011 int-$ per person per year)